Lecture 7

Computing the nullspace of A

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

$$\Big\downarrow \; \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 3R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$$\Big\downarrow \; R_3 \to R_3 - R_2$$

$$U = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

free

Pivot

$$x_1 + 2x_2 + 2x_3 + 2x_4 = 0.$$
$$2x_3 + 4x_4 = 0.$$

Solutions of $Ax=0$: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$ (in $N(A)$)

"special" solutions

Set one free variable $=1$, other free variables $=0$, solve for pivot vars.

$$N(A) = \left\{ s_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

all combinations (=span) of special solutions.

$$x_1 + 2x_2 + 2x_3 + 2x_4 = b_1$$
$$2x_1 + 4x_2 + 6x_3 + 8x_4 = b_2$$
$$3x_1 + 6x_2 + 8x_3 + 10x_4 = b_3$$

$Ax=b$ has no soln if $b_3 \neq b_1 + b_2$.

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 1 \\ 2 & 4 & 6 & 8 & 4 \\ 3 & 6 & 8 & 10 & 5 \end{array}\right]$$

$$\downarrow$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 1 \\ 0 & 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$x_1 + 2x_2 + 2x_3 + 2x_4 = 1$$
$$2x_3 + 4x_4 = 2$$

All soln's to $Ax=b$:

$$x = x_p + x_n$$

$$x_n = s_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Easy way to find x_p: Set free vars $=0$.

$$x_p = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

pivot

free